

No infinite-dimensional Hilbert space admits a norm-compact exhaustion

Candidate full negative answer to the compact-exhaustion proposal
in arXiv:2512.24348

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Status: candidate counterexample/full classification, likely valid. With the standard norm topology, a Hilbert space admits a manifold-style compact exhaustion if and only if it is finite-dimensional. In particular, the compact-exhaustion construction proposed for a nonseparable Hilbert space cannot even be initiated. This does not rule out heat kernels obtained by a different approximation scheme or topology.

1 The source proposal

Jorgensen, Jorgenson, and Smajlovic construct heat kernels on several function spaces from parametrices. In their final open-questions section they recall the classical construction on a noncompact Riemannian manifold: choose open sets Ω_j with compact closures satisfying

$$\overline{\Omega_j} \subset \Omega_{j+1}, \quad \bigcup_{j \geq 1} \Omega_j = M,$$

and pass to a limit of the heat kernels on Ω_j . They then hypothesize that a heat kernel on a nonseparable Hilbert space can similarly be defined through compact exhaustion (arXiv:2512.24348v1, §10.3, p. 34). The complete source page is reproduced next.

The natural topology on a Hilbert space in this analogy is its norm topology. The obstruction below is purely topological and occurs before any analytic choice of Laplacian, boundary condition, or convergence mode.

asymptotic formula in T , as well as the asymptotic growth of the values $\{\phi_n^2(x)\}$. The generalization of these results to the setting of this article now are readily available.

10. FURTHER EXAMPLES AND OPEN QUESTIONS

Let us conclude this article by listing a number of examples. Within many of the examples we will state points which we believe are interesting directions for future studies.

10.1. Finite and infinite graphs. There are many various examples to consider here, so we will list only some of the possibilities.

Suppose $\mathfrak{X}$ is a finite graph, so $\mathcal{B}$ is generated by a finite subsets of singleton sets, each with positive measure. Then, as discussed in [CJKS24] and [JKS26], different examples yield various interesting results. For example, Proposition 19 of [CJKS24] studies a discrete half-line and derives the heat kernel (with a boundary condition) using the heat kernel on the discrete line. Along the way, the proof implies a number of identities associated to the classical I -Bessel function. In [JKS26] it is shown that one can take a Dirac delta function as a parametrix, which yields a known combinatorial formula for the heat kernel on an infinite graph in terms of lengths of certain paths.

10.2. Reproducing kernel Hilbert space. Consider the setting of a reproducing kernel Hilbert space with kernel $H(x, y)$. Then the function $e^{-t}H(x, y)$ is a parametrix.

In general, the function $H(x, y)$ is called the Bergman kernel, and itself is a widely studied function and it possesses “existence and uniqueness” features like the heat kernel, under certain circumstances. Indeed, this is the point of view from Chapter 6 of [Miy06] who is following a method due to Selberg to get (what is called) the Eichler trace formula by constructing the Bergman kernel using a second approach. (For a review of this material using the language of reproducing kernel Hilbert spaces, we refer to unpublished yet available online notes by E. Assing, who refers to the Master’s Thesis of F. Völz.)

In this setting, the small time asymptotics of the heat kernel was used to prove effective sup-norm bounds for the Bergman kernel associated to the space of holomorphic form of fixed weight for finite volume Riemann surfaces; see [FJK19]. The same method of proof has been employed in other settings; see for example [ABR25], [AKvP25] and references therein. It is evident in all of these cases that the proof involves general properties of heat kernels, and is widely adaptable under general circumstances.

There is a very general construction of positive definite kernels, which has many applications; see section 6 of [JT21]. Namely, let $\{f_n\}$ be an orthonormal basis of the separable Hilbert space $\mathfrak{X}$ under study, finite or not, and let $\{Z_n\}$ be an i.i.d. system of standard Gaussians. For every $x \in \mathfrak{X}$, set $G_x = \sum f_n(x)Z_n$. Then the positive definite kernel $K(x, y) = \sum f_n(x)f_n(y)$ can be realized as $K(x, y) = \mathbb{E}(G_x G_y)$, where $\mathbb{E}$ denotes the expected value. We refer to [JT21], specifically Theorem 6.3, and references therein for further discussion and applications. The results in the present article now provide an explicit construction of the corresponding heat kernel, thus makes available all the corresponding mathematical tools for future studies.

10.3. Non-separable Hilbert spaces. In the setting of Riemannian geometry, one can define a heat kernel on a non-compact space through compact exhaustion; see, for example, Chapter VIII of [Cha84], specifically page 188. If M is a noncompact Riemannian manifold, one considers a sequence

Ω_j for integers $j \geq 1$ whose compactification $\bar{\Omega}_j$ lies in Ω_{j+1} and such that $\bigcup_{j=1}^{\infty} \Omega_j = M$. Then one

defines the heat kernel on M to be the limit in j of the heat kernels on Ω_j . We hypothesize that similarly one can define a heat kernel on a nonseparable Hilbert space through compact exhaustion. Again, we leave that study for a future investigation.

2 Exact classification

Call a sequence $(\Omega_j)_{j \geq 1}$ a *norm-compact exhaustion* of a Hilbert space $\mathcal{H}$ if every Ω_j is norm open, $\overline{\Omega_j}$ is norm compact, $\overline{\Omega_j} \subset \Omega_{j+1}$, and $\bigcup_j \Omega_j = \mathcal{H}$. The nesting is not needed for the negative result.

Theorem 1. *For a real or complex Hilbert space $\mathcal{H}$, the following are equivalent.*

1. $\mathcal{H}$ admits a norm-compact exhaustion.
2. $\mathcal{H}$ is a countable union of norm-compact subsets.
3. $\mathcal{H}$ is finite-dimensional.

Proof. The first condition implies the second because $\mathcal{H} = \bigcup_j \overline{\Omega_j}$.

Suppose $\mathcal{H} = \bigcup_{j \geq 1} K_j$ with each K_j norm compact. If $\mathcal{H}$ is infinite-dimensional, every K_j has empty interior. Indeed, if $B(x, r) \subset K_j$ for some $r > 0$, then compactness and closedness give $\overline{B}(x, r) \subset K_j$. Translation by $-x$ and dilation by $1/r$ would make the closed unit ball of $\mathcal{H}$ compact. This is impossible in an infinite-dimensional normed space: Riesz’s lemma supplies a sequence of unit vectors with pairwise distance bounded below, hence with no convergent subsequence. Thus each compact K_j is closed and nowhere dense.

But a Hilbert space is complete, so the Baire category theorem says that it cannot be a countable union of closed nowhere-dense sets. Hence $\mathcal{H}$ must be finite-dimensional.

Conversely, if $\dim \mathcal{H} < \infty$, let $\Omega_j = B(0, j)$. Heine–Borel gives compactness of $\overline{B}(0, j)$, and $\overline{B}(0, j) \subset B(0, j+1)$ gives the required strict nesting. $\square$

Corollary 2. *No nonseparable Hilbert space has the norm-compact exhaustion required by the proposal in arXiv:2512.24348. More strongly, no infinite-dimensional Hilbert space has one, separable or not.*

There is also a shorter obstruction tailored to the stated nonseparable case. Every compact metric space is separable, and a countable union of separable subspaces is separable. Therefore a nonseparable metric space can never be σ -compact. This argument needs neither linear structure nor completeness; Theorem 1 gives the sharper finite-dimensional classification.

3 Why the manifold analogy breaks

The Riemannian construction uses local compactness. Every point lies in a relatively compact coordinate neighborhood, and second countability lets one select a countable exhaustion. By contrast, a normed space is locally compact exactly when it is finite-dimensional. The proof above is the global Baire-category form of the same obstruction.

Consequently, there are no norm-open domains Ω_j with compact norm closures that cover an infinite-dimensional Hilbert space. In particular, there are no domain heat kernels of the proposed kind whose limit in j could define the desired kernel. The conjectured procedure fails at its first geometric input, independently of every heat-kernel estimate.

4 Deep upgrades and alternative readings

Sequential separable approximations. The nonseparable obstruction persists after relaxing “compact” to “separable.” If (S_j) is any sequence of separable subsets of a metric space, then $\overline{\bigcup_j S_j}$

is separable. Hence no sequence of separable subspaces, and in particular no sequence of finite-dimensional subspaces, can have dense union in a nonseparable Hilbert space. A finite-dimensional Galerkin sequence therefore cannot replace the proposed compact exhaustion globally.

Weak topology. Because Hilbert spaces are reflexive, the closed norm balls are weakly compact and their union is the whole space. This is a legitimate weakly compact cover, but not the stated manifold-style exhaustion. In an infinite-dimensional Hilbert space every weak neighborhood contains an unbounded affine subspace: finitely many defining functionals have a common nonzero kernel direction. Thus bounded balls have empty weak interior, and one does not have $K_j \subset \text{int}_{\text{weak}} K_{j+1}$ or relatively compact weak neighborhoods. More importantly, norm-topology boundary conditions, local differential operators, and convergence of heat kernels do not follow from weak compactness. A weak-topology construction would be a different problem requiring new definitions and estimates.

Nets. An uncountable directed family of compact sets can cover any space (singletons already do), but this loses the sequential exhaustion and canonical limit used in the source analogy. In a nonseparable Hilbert space, nets of finite-dimensional subspaces may be directed by finite subsets of an orthonormal basis. Turning such a net into compatible heat kernels and proving a topology-independent limit would again be a new construction, not the proposed compact-exhaustion method.

5 Conclusion, scope, and checks

The source’s compact-exhaustion hypothesis has a definitive negative answer in the norm topology: the method is available precisely for finite-dimensional Hilbert spaces. The result is stronger than the nonseparable case requested and is sharp.

It does *not* assert that nonseparable Hilbert spaces carry no heat semigroup, no transition kernel relative to additional measurable data, or no useful finite-dimensional net. It says that the specific Riemannian compact-exhaustion mechanism cannot be transferred to an infinite-dimensional Hilbert space with its norm topology.

Upgrade log. The first route used the immediate separability obstruction. A second route upgraded it to the exact finite-dimensional classification by Baire category and Riesz’s lemma. A third checked that sequential separable/Galerkin exhaustions also cannot be dense in the nonseparable case. A fourth isolated the weak-topology and net reinterpretations and showed why neither is the claimed manifold analog.

Novelty check. Bounded primary-source searches through 11 August 2026 used the exact source title and phrase together with “nonseparable Hilbert space,” “compact exhaustion,” and “heat kernel.” They recovered the source and classical manifold compact-exhaustion uses, but no later primary source addressing this proposal. The Baire/Riesz classification is elementary classical topology; the candidate contribution is its application as a sharp correction to the new heat-kernel proposal, not novelty of those underlying lemmas.

Human review. Confirm that “compact exhaustion” is intended in the norm topology, as in the displayed Riemannian analogy. Under that reading the obstruction and classification are unconditional.